



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to MAHALAKSHMI TRADERS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

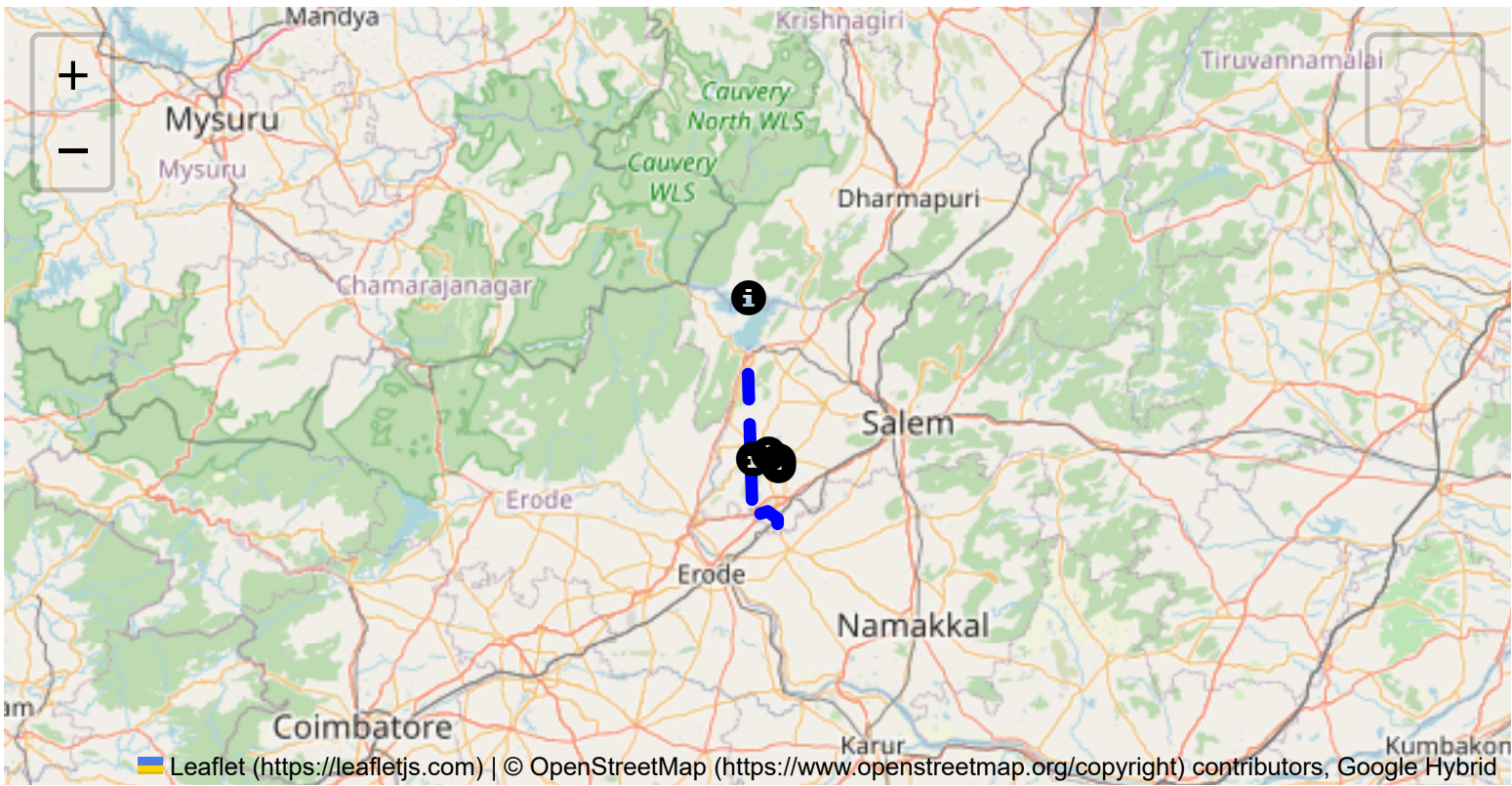
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

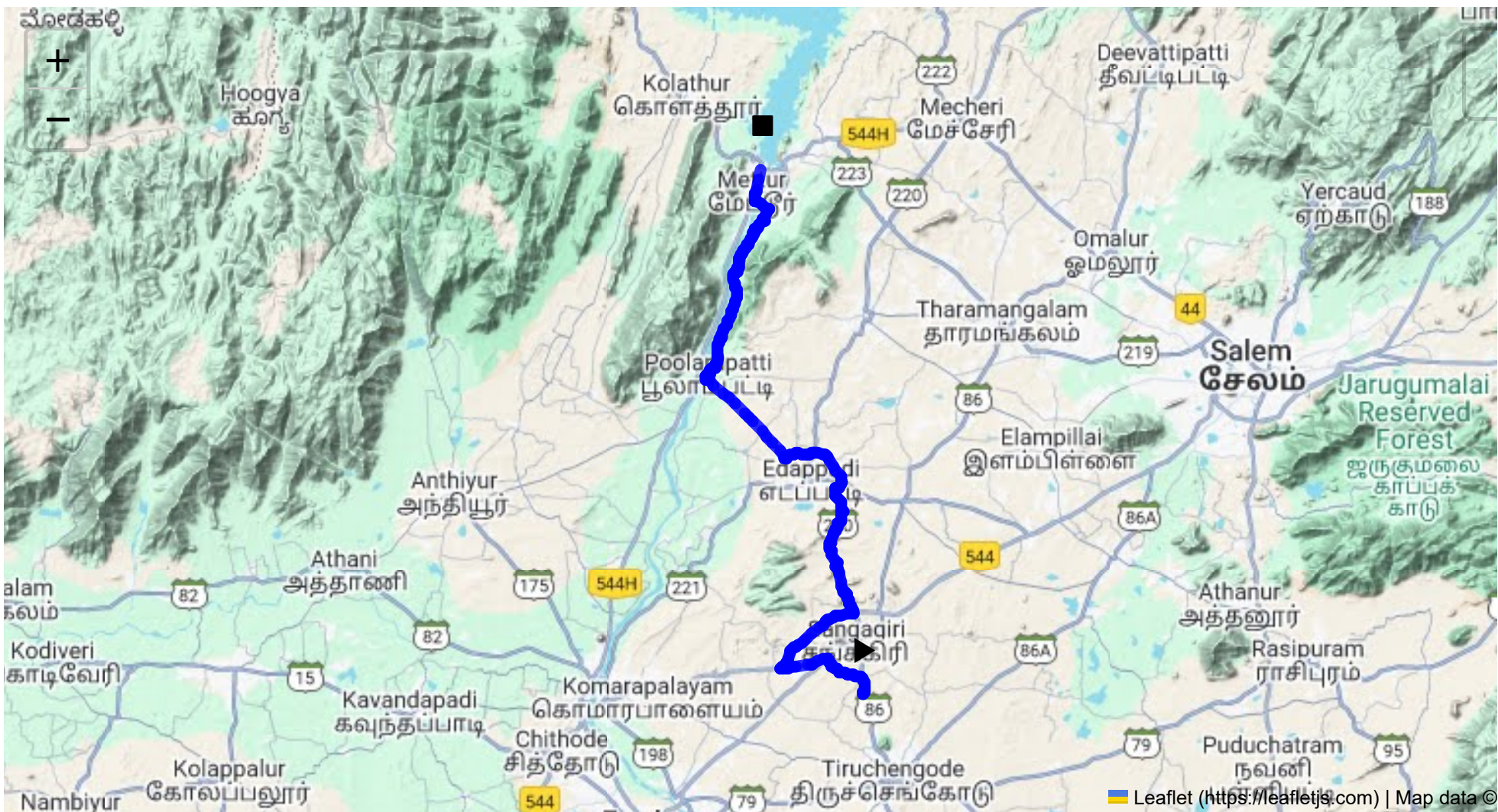
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 60.04 km
Estimated Duration: 1.4 hours
Adjusted Duration (Heavy Vehicle): 1.8 hours
Start: (11.4381, 77.8734)
End: (11.797656, 77.802506)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route from CVQF+23W Sangagiri to QRX2+2XV Mettur spans approximately 60 kilometers and passes through areas such as Sankari RS, Sankari West, and Padaiveedu. It mainly traverses state highways and local roads, providing a vital link between these regions. The terrain is mostly flat with occasional elevated sections.

2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu usually exhibits a tropical climate with heavy rainfall during the northeast monsoon (October to December). During this period, heavy rains can lead to waterlogging and slippery roads. Excessive heat in the summer months (April to June) can cause road surface damage and discomfort.

3. Traffic Patterns and Congestion-Prone Areas

- **Peak Hours:** Usually, morning (8:00-10:00 AM) and evening (5:00-8:00 PM) are the peak traffic times.
- **Congestion-Prone Areas:** The areas near major junctions such as Sankari RS and Bhavani Main Road may experience congestion, especially during peak hours.

4. Assessment of Road Quality and Infrastructure

- **Road Quality:** Most sections are well-paved but expect occasional potholes and uneven surfaces, particularly in rural areas.
- **Infrastructure:** Adequate signage and lighting are present on highways, but local roads may lack sufficient lighting.

5. Suggestions for Alternative Routes for Emergencies

In the event of roadblocks or accidents, consider bypassing via smaller village roads linking to adjacent state highways. Ensure these routes are checked for suitability, as they might not be capable of handling heavy vehicles under normal conditions.

6. Summary of Local Regulations Affecting Hazardous Material Transport

Heavy vehicles carrying hazardous materials are required to adhere to strict timing regulations, often restricted to non-peak hours to minimize risk. Permits should be up to date, and routes often reviewed to avoid populated areas when possible.

7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials

- **Collisions/Accidents:** Historical data may show incidents at key junctions, often related to speeding or poor weather conditions.
- **Material Spills:** Rare but documented instances often attributed to mechanical failure or driver error.

8. Environmental Considerations and Sensitive Areas

The route may pass near agricultural zones and water bodies, requiring extra caution to prevent contamination from spills or accidents. No significant wildlife sanctuaries along this specific route, but local flora can be affected.

9. Analysis of Communication Coverage

Mobile network coverage is generally good in populated areas but sparse along remote stretches. Communication dead zones may be experienced in more rural sections between major towns.

10. Estimated Emergency Response Times for Different Route Segments

- **Urban Areas:** Potential response time is around 15-30 minutes.
- **Rural Sections:** Due to distance and road quality, response times can extend to 45 minutes or more.

12. Overall Summary of Risk Assessment

This route is moderately safe for transporting hazardous materials, with key points of concern being weather-related hazards during monsoons, peak hour traffic congestion, and occasional infrastructure limitations on local roads. Proactive measures, including route familiarization, up-to-date communication equipment, and contingency planning for emergencies, are recommended for safe transit.

By observing these elements and maintaining a proactive approach towards safety management and regulation compliance, the risk associated with transporting hazardous materials on this route can be effectively mitigated.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.58300, 77.85813	15 KM/Hr
1	Roundabout	High	11.60019, 77.81968	15 KM/Hr
2	Roundabout	High	11.77576, 77.79789	15 KM/Hr
3	Turn	Medium	11.43822, 77.87348	30 KM/Hr
4	Turn	High	11.43968, 77.87345	15 KM/Hr
5	Turn	High	11.44029, 77.87544	15 KM/Hr
6	Turn	Medium	11.44900, 77.87406	30 KM/Hr
7	Turn	Medium	11.45357, 77.85789	30 KM/Hr
8	Turn	High	11.45352, 77.85698	15 KM/Hr
9	Turn	Medium	11.45847, 77.85140	30 KM/Hr
10	Turn	Medium	11.46303, 77.84945	30 KM/Hr
11	Turn	Medium	11.46314, 77.84928	30 KM/Hr
12	Turn	High	11.45542, 77.81725	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
13	Turn	High	11.45634, 77.81673	15 KM/Hr
14	Blind Spot	Blind Spot	11.49397, 77.86740	10 KM/Hr
15	Turn	Medium	11.49404, 77.86724	30 KM/Hr
16	Turn	Medium	11.50866, 77.86053	30 KM/Hr
17	Turn	Medium	11.52587, 77.85524	30 KM/Hr
18	Turn	Medium	11.54750, 77.85273	30 KM/Hr
19	Turn	Medium	11.55039, 77.85511	30 KM/Hr
20	Turn	Medium	11.55481, 77.85639	30 KM/Hr
21	Turn	High	11.57893, 77.85430	15 KM/Hr
22	Turn	Medium	11.57903, 77.85441	30 KM/Hr
23	Turn	Medium	11.58022, 77.85744	30 KM/Hr
24	Turn	High	11.65398, 77.76444	15 KM/Hr
25	Turn	High	11.65608, 77.76465	15 KM/Hr
26	Turn	High	11.65623, 77.76451	15 KM/Hr
27	Turn	Medium	11.65760, 77.76513	30 KM/Hr
28	Turn	High	11.65764, 77.76613	15 KM/Hr
29	Turn	High	11.66103, 77.76757	15 KM/Hr
30	Turn	High	11.66090, 77.76849	15 KM/Hr
31	Turn	Medium	11.66123, 77.76890	30 KM/Hr
32	Turn	High	11.66122, 77.76992	15 KM/Hr
33	Turn	High	11.66386, 77.77083	15 KM/Hr
34	Turn	High	11.66386, 77.77125	15 KM/Hr
35	Turn	Medium	11.66400, 77.77136	30 KM/Hr
36	Turn	High	11.66857, 77.77231	15 KM/Hr
37	Turn	Medium	11.66873, 77.77191	30 KM/Hr
38	Turn	High	11.66890, 77.77178	15 KM/Hr
39	Turn	Medium	11.66947, 77.77209	30 KM/Hr
40	Turn	High	11.72628, 77.78415	15 KM/Hr
41	Turn	High	11.72745, 77.78411	15 KM/Hr
42	Turn	Medium	11.72759, 77.78433	30 KM/Hr
43	Turn	Medium	11.72751, 77.78550	30 KM/Hr
44	Turn	Medium	11.72785, 77.78582	30 KM/Hr
45	Turn	Medium	11.73941, 77.78811	30 KM/Hr
46	Turn	Medium	11.73971, 77.78838	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
47	Turn	Medium	11.74000, 77.78938	30 KM/Hr
48	Turn	Medium	11.75274, 77.79710	30 KM/Hr
49	Turn	Medium	11.75599, 77.79872	30 KM/Hr
50	Turn	High	11.75676, 77.79842	15 KM/Hr
51	Turn	High	11.75724, 77.79860	15 KM/Hr
52	Turn	Medium	11.75752, 77.79996	30 KM/Hr
53	Turn	Medium	11.76229, 77.80500	30 KM/Hr
54	Turn	High	11.76312, 77.80533	15 KM/Hr
55	Turn	Medium	11.76347, 77.80512	30 KM/Hr
56	Turn	High	11.76402, 77.80425	15 KM/Hr
57	Turn	Medium	11.76436, 77.80420	30 KM/Hr
58	Turn	Medium	11.76990, 77.80761	30 KM/Hr
59	Turn	High	11.77028, 77.80821	15 KM/Hr
60	Turn	High	11.77045, 77.80821	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
2	hospital	S.K.A Hospital	11.7816111, 77.797688	30 km/h	Medium
3	hospital	Sathya Hospital	11.7864589, 77.8004201	30 km/h	Medium
4	clinic	Saraswathi Poly Clinic	11.7905491, 77.8007345	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
1	college	Govt. Arts College, Mettur	11.7756806, 77.7958483	30 km/h	Medium
5	school	Govt. Hr. Sec. School	11.792309, 77.8006337	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.58300, 77.85813



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.60019, 77.81968



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.77576, 77.79789



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45357, 77.85789



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45352, 77.85698



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45847, 77.85140



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46303, 77.84945



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46314, 77.84928



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45542, 77.81725



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45634, 77.81673



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.49397, 77.86740



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49404, 77.86724



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.50866, 77.86053



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.52587, 77.85524



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.54750, 77.85273



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.55039, 77.85511



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.55481, 77.85639



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.57893, 77.85430



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.57903, 77.85441



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.58022, 77.85744



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.65398, 77.76444



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.65608, 77.76465



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.65623, 77.76451



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.65760, 77.76513



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.65764, 77.76613



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.66103, 77.76757



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.66090, 77.76849



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.66123, 77.76890



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.66122, 77.76992



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.66386, 77.77083



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.66386, 77.77125



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.66400, 77.77136



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.66857, 77.77231



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.66873, 77.77191



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.66890, 77.77178



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.66947, 77.77209



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.72628, 77.78415



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.72745, 77.78411



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.72759, 77.78433



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.72751, 77.78550



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.72785, 77.78582



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.73941, 77.78811



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.73971, 77.78838



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.74000, 77.78938



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.75274, 77.79710



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.75599, 77.79872



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.75676, 77.79842



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.75724, 77.79860



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.75752, 77.79996



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.76229, 77.80500



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.76312, 77.80533



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.76347, 77.80512



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.76402, 77.80425



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.76436, 77.80420



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.76990, 77.80761



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.77028, 77.80821



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.77045, 77.80821

